Downstream Utility of Synthetic Data for Clustering

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Abstract

Synthetic data is increasingly released as a privacy-preserving substitute for confidential data, but its value rests on whether an analyst reaches the same conclusions from the synthetic release as from the original records. We evaluate that premise for cluster analysis, a task whose output depends almost entirely on the geometry of the point cloud and which therefore probes exactly the higher-order structure that marginal resemblance checks cannot see. Because a recovered partition responds to local density, dispersion and separability at once, the comparison serves two purposes: it measures analytical utility for a specific downstream task, and it doubles as a demanding test of resemblance between the synthetic and the original data. Within a utility-discrepancy framework we compare a partitioning method ($\boldsymbol{k}$-means) and a hierarchical method (Ward's linkage) on data generated by sequential CART synthesis, across a factorial design of 144 scenarios crossing the number of groups ($\boldsymbol{k \in \{2, 3, 4\}}$), separation ($\boldsymbol{\sigma \in \{0.1, 2, 6, 10\}}$), dimensionality ($\boldsymbol{p \in \{2, 5, 10\}}$), feature correlation ($\boldsymbol{\rho \in \{0, 0.4\}}$) and distribution family (Normal, Gamma), yielding 720,000 synthetic-versus-original evaluations. Our primary metric is the proportion of evaluations in which the number of clusters recovered from the synthetic data matched the number recovered from the original data. Separation dominates this agreement, which rises from **0.75** in the overlapping regime to **0.98** at $\boldsymbol{\sigma = 10}$; agreement degrades as $\boldsymbol{k}$ and $\boldsymbol{p}$ grow. Measured against the generating truth instead, the two algorithms are statistically indistinguishable (**0.585** versus **0.585**; paired Wilcoxon $\boldsymbol{p = 0.39}$; **92%**

agreement), and they differ systematically in the geometric bias they impose: $\boldsymbol{k}$-means is the more conservative estimator on centroid location and dispersion, while its within-cluster sum-of-squares objective suppresses the size imbalance that skewed groups would otherwise produce. CART is close to utility-neutral for spherical Normal components and measurably degrades recovery for skewed Gamma components, with distortion growing in $\boldsymbol{p}$. We conclude that synthetic-data utility for clustering is a property of the generator–algorithm–design triple and must be validated against the intended analysis.

Keywords: Synthetic data, Cluster analysis, Statistical disclosure control, Analytical utility, Simulation study

1 Introduction

Synthetic data are artificial records produced by a generative model fitted to a confidential dataset, released in place of the original records so that the statistical structure of the data survives while no real individual is exposed. The idea originates in statistical disclosure control, where Rubin (1993) proposed treating the confidential values as missing and releasing multiply-imputed replacements, and Little (1993) developed the companion analysis of masked data; Drechsler (2011) gives the book-length treatment of the resulting framework, and Matthews and Harel (2011) survey the wider family of disclosure-limitation methods within which synthesis sits. What began as a mechanism for statistical agencies has since spread well beyond official statistics, and general-purpose synthesisers are now distributed as statistical software (Nowok et al, 2016; Templ et al, 2017), with administrative-data research an active area of application (Kokosi et al, 2022). Synthetic records are now generated for health services research (Gonzales et al, 2023; Giuffrè and Shung, 2023), for biobehavioural science, where they also serve reproducibility and hypothesis generation (Quintana, 2020), and, increasingly, through deep generative architectures applied to tabular data (Goyal and Mahmoud, 2024; Zhao et al, 2024).

Cluster analysis occupies the opposite end of the analytical pipeline. It is the canonical unsupervised task: given unlabelled observations, partition them into groups that are internally homogeneous and mutually distinct. Two paradigms have dominated applied practice for decades. Centroid-based partitioning, of which k-means (MacQueen, 1967) is the archetype, minimises the within-cluster sum of squares and remains, by a wide margin, the most widely deployed clustering algorithm (Ahmed et al, 2020). Connectivity-based agglomeration, typified by Ward's linkage (Ward, 1963), instead builds a dendrogram by successive merging and imposes no explicit convexity assumption. Both remain central to contemporary applications: in single-cell transcriptomics, where partitions define cell types (Zappia and Oshlack, 2018), and in mass cytometry, where the choice among clustering methods materially changes the biological conclusion (Liu et al, 2019).

These two literatures meet whenever a clustering analysis must be performed on data that cannot be released in raw form. The motivation is partly privacy, because clinical and administrative records are the very datasets on which subgroup discovery is most valuable and least permissible (Gonçalves et al, 2020; El Emam et al, 2020; Bhanot et al, 2021), and partly practical, since synthetic records also serve as an augmentation and model-development resource when access to the originals is slow or restricted (Rankin et al, 2020). In either case the analyst runs an unsupervised procedure on data they did not observe, and the question of whether the resulting partition would have been obtained from the original records is not rhetorical.

That question is the concern of *analytical utility*. It is conventional to distinguish *general* utility, which asks whether the synthetic data resemble the original in some global sense, from *specific* utility, which asks whether a particular analysis yields matching conclusions (Snoke et al, 2018). The distinction matters because the two can come apart: a generator that reproduces marginal distributions and low-order correlations faithfully may still distort the higher-order geometry on which a specific procedure depends. Empirical work bears this out. Benchmarking studies of synthetic health records find that utility rankings depend heavily on which downstream task is used to measure them (Yan et al, 2022; El Emam et al, 2022), and direct replications of published analyses on synthetic counterparts show agreement that varies by study and by estimand (Reiner Benaim et al, 2020). A recent systematic review of tabular synthesis tools makes the same point from the measurement side, finding no consensus on which utility metrics should be reported and little agreement between them (Lautrup et al, 2024). Utility is therefore not a scalar property of a dataset; it is a property of the pairing between a dataset and the analysis applied to it.

Clustering is a demanding probe of that pairing. Unlike a regression coefficient, which summarises a conditional mean and is comparatively robust to local perturbation, a partition is sensitive to the fine structure of the point cloud: relative positions, local density, separability, within-group dispersion. Two datasets can share identical means, variances and pairwise correlations and still yield entirely different partitions. Any distortion a generator introduces into local geometry should therefore surface in a clustering analysis before it surfaces in a resemblance check.

This article asks whether CART-based sequential synthesis, the default non-parametric engine of `synthpop` (Nowok et al, 2016) and the method most likely to be used in practice on tabular data, preserves the structure that clustering relies on. It extends the line of inquiry opened by Chen (2025), which established a simulation harness for this question but restricted attention to k-means on Normal data and relied on a resemblance statistic that proved insensitive to structural quality. We broaden the scope in three directions: a second, structurally different clustering paradigm; a skewed distribution family that exposes the axis-aligned behaviour of tree-based synthesis; and a battery of ground-truth-referenced structural metrics that do not depend on the synthetic data alone.

In this article we ask three connected questions. First, whether CART-synthesised data retains the signal needed to recover the number of clusters, and how that depends on separation, cluster count, dimensionality, feature correlation and distribution family. Second, whether the geometry survives beyond the count: how much fidelity of centroid location, within-group dispersion and group balance is lost under synthesis, and which design factors drive that loss. Third, whether partitioning and hierarchical methods respond to synthetic distortion alike, or whether one is systematically more robust than the other. We answer them with a factorial simulation in which every scenario is synthesised many times over, so that each comparison is an average over replicate generators rather than a single draw, and we report the number of clusters against two references: the groups planted by the simulation, and the groups a clustering algorithm recovers from the original data.

2 Methods

The study is a controlled factorial simulation. We first state the two clustering algorithms, then the generation of reference and synthetic data, then the metrics used to quantify structural degradation, and finally the way those metrics are summarised across replications.

2.1 A discrepancy framework for structural utility

Let D_R denote the reference data and G a generative mechanism, so that $D_S = G(D_R)$ is the synthetic dataset. Let A be the downstream algorithm, producing summaries $\theta_R = A(D_R)$ and $\theta_S = A(D_S)$. Utility is quantified by a discrepancy

$$\Delta(\theta_R, \theta_S) = f(N, p, \rho, \sigma, k), \tag{1}$$

studied as a function of the sample size N, dimensionality p, feature correlation ρ, separation σ and number of latent groups k. A generator is useful for a task over a region of design space when Δ remains small throughout it; the scientific content of an evaluation is the shape of Δ: which factors inflate it, which leave it unchanged, and whether different algorithms respond alike.

In the clustering setting, θ comprises the estimated number of groups $\hat{k}$ together with the geometry of the recovered partition: its centroids, its group sizes and its within-group dispersion. Accordingly Δ is instantiated as a family of structural-fidelity measures defined in Section 2.4.

To isolate the mechanism we use simulations with known ground truth rather than real administrative data. Each scenario fixes a configuration $(N, p, k, \rho, \sigma, \text{distribution})$ for which the true group membership of every point is known by construction, so recovery is scored against ground truth rather than against the synthetic data itself. This avoids the circularity that undermines resemblance-only evaluations, and the factorial structure allows Δ in Eq. (1) to be traced as an explicit function of each design factor.

2.2 Clustering algorithms

We compare two standard paradigms: k-means (MacQueen, 1967) as a partitioning method, and agglomerative clustering with Ward's linkage (Ward, 1963) as a hierarchical one. They are chosen because they embody different geometric assumptions and should therefore respond differently to the artifacts introduced by tree-based synthesis. k-means minimises the within-cluster sum of squares and so presumes convex, isotropic, comparably sized groups, with decision boundaries given by the bisectors of the centroid Voronoi tessellation; we retain the lowest-inertia solution over ten random initialisations. Ward's linkage merges the pair of clusters whose union least increases the total within-cluster error sum of squares, makes no explicit convexity assumption and can follow connected structure, but is correspondingly more exposed to spurious local density. Appendix A states both objectives formally.

2.3 Data generation

2.3.1 Original data

Each original dataset D_R contains $N = 1000$ observations drawn from a k-component mixture in $\mathbb{R}^p$ with known group labels. Component centroids are placed so that the minimum pairwise centroid distance equals a controlled separation σ: the first centroid sits at the origin and each subsequent centroid is positioned at distance at least σ from those already placed. Within-component covariance is governed by a correlation matrix with off-diagonal entries ρ, giving spherical components at $\rho = 0$ and correlated components at $\rho = 0.4$. In the *Normal* family components are multivariate Gaussian with the specified covariance and centroid. In the *Gamma* family, skewed components (shape $= 1.5$, rate $= 5$) are scaled, given the target correlation through a Cholesky factor, and shifted to the component centroid, producing curved, right-skewed groups that stress the axis-aligned nature of CART.

The factors we vary are the same high-level geometric properties used to parameterise cluster-analysis benchmarks generally, namely overlap between groups, their number, the ambient dimension and the shape of the components (Zellinger and Bühlmann, 2025); here they serve not to benchmark clustering algorithms against each other but to locate the regions in which synthesis preserves the structure those algorithms depend on.

Table 1 lists the design factors. Each factor is varied to isolate a distinct hypothesis about when synthesis should succeed or fail: crossing all five gives $3 \times 4 \times 3 \times 2 \times 2 = 144$ scenarios. Each scenario is replicated across $n = 5$ independent original datasets, and each original dataset is synthesised $m = 1000$ times, so the executed design comprises $144 \times 5 \times 1000 = 720{,}000$ synthetic-versus-original evaluations.

2.3.2 Synthetic data (CART)

For every original dataset, synthetic data $D_S = G(D_R)$ are generated with the CART engine of `synthpop` (Nowok et al, 2016), using proper synthesis and a minimum leaf size of ten. Variables are synthesised sequentially: the first is sampled from its observed marginal, and each subsequent variable X_j is modelled by a regression tree (Breiman et al, 1984) on the already-synthesised variables, with synthetic values drawn from the empirical leaf distribution,

$$X_j^{\text{syn}} \sim \widehat{P}\big(X_j \mid \text{leaf}(X_1^{\text{syn}}, \dots, X_{j-1}^{\text{syn}})\big). \tag{2}$$

This makes the method flexible and distribution-free, but it imposes an axis-aligned, piecewise-constant structure on the joint distribution: the synthetic point

Table 1 Factors of the simulation design and the question each addresses. Crossing all five gives 144 scenarios; with $n = 5$ original replicates and $m = 1000$ synthetic draws each, the design yields 720,000 evaluations.

Factor	Levels	Rationale
Separation σ	0.1, 2, 6, 10	Spans overlapping to clearly distinct groups; tests how much signal must be present before synthesis can preserve it.
Number of groups k	2, 3, 4	More groups means more boundaries the generator must reproduce.
Dimensionality p	2, 5, 10	Each synthesised dimension compounds the axis-aligned leaf structure of the generator.
Correlation ρ	0, 0.4	Tests whether dependence between features degrades a variable-by-variable synthesis.
Distribution family	Normal, Gamma	Contrasts smooth spherical density with skewed, curved density.
Sample size N	1000 (fixed)	Held constant so that variation is attributable to the factors above.

cloud is assembled from rectangular blocks whose boundaries are the tree splits. When the original data are smooth and spherical, these blocks tile the space finely enough to be invisible; when the data are curved, skewed or heavy-tailed, they become visible as a hardening of smooth density gradients into discrete steps. Detecting precisely this failure mode, and measuring how strongly it propagates into the recovered partition, is the purpose of the evaluation. The ground-truth group label is carried through synthesis so that the true partition remains available as a reference.

Synthesis was carried out with `synthpop` 1.9.2 (Nowok et al, 2016) under R 4.3 (R Core Team, 2024), and clustering and evaluation with `scikit-learn` (Pedregosa et al, 2011) in Python. Both stages are embarrassingly parallel over scenarios and were distributed across 18 worker processes on a single multi-core workstation.

2.4 Evaluation metrics

Structural utility is quantified by one detection metric and three geometric-fidelity metrics. The fidelity metrics require the groups recovered in D_S to be matched to those in D_R, since cluster labels are arbitrary. We build the matrix of pairwise centroid distances between the k original and k synthetic clusters and solve the resulting linear assignment problem (Kuhn, 1955), obtaining the one-to-one matching that minimises total centroid distance. All fidelity metrics are computed on the aligned clusters.

Recovery of the number of groups.

For each dataset the number of groups is estimated by maximising the average silhouette width (Rousseeuw, 1987) over candidates $\hat{k} \in \{2, \ldots, k_{\max}\}$, with $k_{\max} = \min(k+3, 8)$; the supplementary contingency analysis instead uses a fixed $k_{\max} = 8$ so that the two data sources are searched over an identical range. A trial succeeds when

$\hat{k} = k$, and for a configuration with M evaluations the recovery rate is

$$\mathrm{RR} = \frac{1}{M} \sum_{m=1}^{M} \mathbb{1}\big[\hat{k}_m = k\big]. \tag{3}$$

Because $\hat{k}$ is scored against the construction-time ground truth rather than against the synthetic data, the metric is free of the circularity that affects resemblance-only evaluations.

Group-balance error (ΔG).

The Gini coefficient of the cluster-size vector measures how balanced a partition is. For sizes $\{n_1, \ldots, n_k\}$,

$$G = \frac{\sum_{a=1}^{k} \sum_{b=1}^{k} |n_a - n_b|}{2k \sum_{a=1}^{k} n_a}, \tag{4}$$

with $G = 0$ for perfectly equal groups. The fidelity metric is $\Delta G = |G_S - G_R|$.

Mean centroid distance (MCD).

With aligned original and synthetic centroids μ_j^R and μ_j^S,

$$\mathrm{MCD} = \frac{1}{k} \sum_{j=1}^{k} \big\|\mu_j^R - \mu_j^S\big\|, \tag{5}$$

so small values indicate that synthesis preserves the location of the group structure.

Mean variance difference (MVD).

Writing v_j^R and v_j^S for the mean squared distance of points to their centroid in aligned original and synthetic clusters,

$$\mathrm{MVD} = \frac{1}{k} \sum_{j=1}^{k} \big|v_j^R - v_j^S\big|, \tag{6}$$

capturing whether groups retain their characteristic tightness or spread.

All variables are standardised before clustering. Crucially, the scaler is fit on the original data and applied unchanged to the corresponding synthetic dataset, so that MCD and MVD are expressed in a common coordinate system; standardising each dataset separately would rescale away part of the very distortion the metrics are meant to detect (see Section 4.2).

Recovery rates are proportions over evaluations and are reported directly. Fidelity metrics are aggregated first to a scenario mean and then averaged across scenarios, with dispersion reported as a 95% interval computed from the scenario-level means; using the scenario as the unit of replication avoids treating the m synthetic draws from one original dataset as independent. Algorithm comparison uses a paired Wilcoxon signed-rank test across the 144 scenario-level means. Because estimating the number of groups requires a silhouette scan over candidate values for each algorithm, and the geometric metrics require additional fixed-k fits, the executed design amounts to several million clustering fits; the geometric-fidelity metrics additionally require a Hungarian alignment per pair.

For visualisation we use the two-dimensional scenarios directly, plotting the standardised features against each other; no dimensionality reduction is applied, so the panels show the geometry the algorithms actually operate on. In the paired original/synthetic displays, cluster labels are matched by the same Hungarian alignment used for the fidelity metrics and then oriented so that colour tracks vertical position consistently across panels.

3 Results

3.1 The block-like artifact

Figures 1 and 2 show a two-dimensional, two-group scenario at moderate separation ($\sigma = 2$): the original data in the top row, one CART synthetic replicate below, each partitioned by both algorithms. Reading across a row shows how the two algorithms divide the same cloud; reading down a column shows what synthesis did to that division.

On Normal data (Fig. 1) the two compact masses survive synthesis. The synthetic cloud is visibly sparser in the tails and shows faint horizontal striping where leaf boundaries fall, but both algorithms cut it in essentially the same place as they cut the original, and the recovered groups occupy the same regions. The left column also illustrates why this scenario is hard for a different reason: at $\sigma = 2$ the true groups overlap substantially, so neither algorithm reproduces the true labelling exactly: the boundary they find is a clean vertical split of the combined cloud rather than the diagonal boundary between the generating components.

On Gamma data (Fig. 2) the mechanism becomes visible. The original groups are curved and right-skewed, with a dense corner and long tails running out along both axes. In the synthetic cloud those smooth tails harden into discrete, axis-aligned patches, and the dense corner is reproduced as a set of rectangular blocks rather than a continuous gradient. This is the signature artifact of sequential tree synthesis: the piecewise-constant leaf structure of Eq. (2) becomes apparent whenever the underlying density is not smooth and spherical. The partitions still divide the cloud in roughly the same orientation, but the boundary is displaced and the resulting groups have different internal spread, the distortion that the fidelity metrics quantify below.

3.2 Number of groups

The number of clusters recovered from synthetic data can be judged against two references, and they answer different questions. Comparing it with the number of groups planted by the simulation asks whether the synthetic data supports recovery of the underlying truth, and so confounds two failures: distortion introduced by synthesis, and the inability of the algorithm to find the planted structure at all. Comparing it

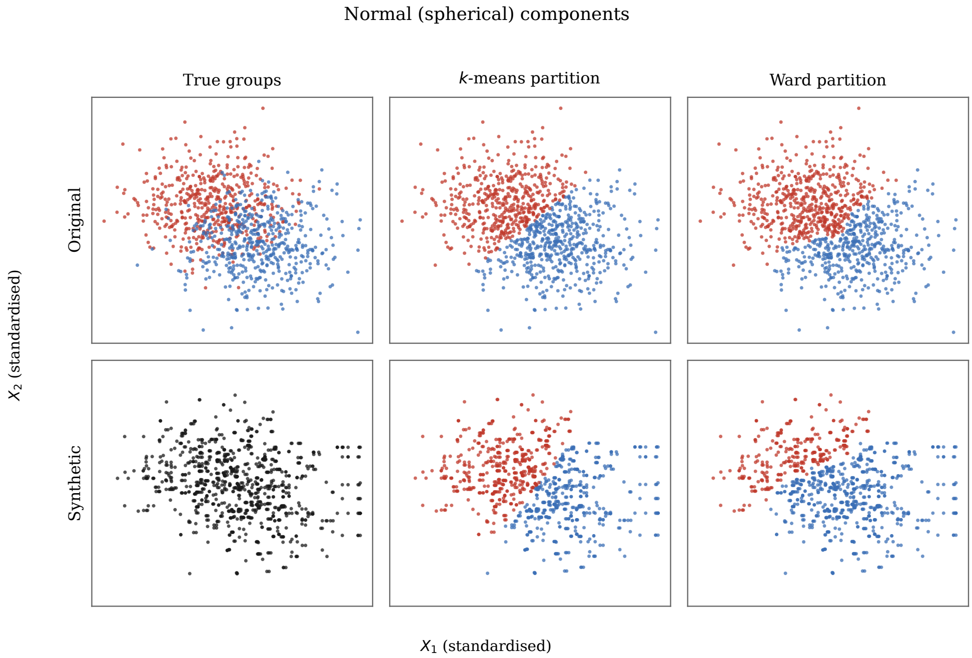

Fig. 1 Normal (spherical) components: geometry largely preserved under synthesis. Rows are the original data (top) and one CART synthetic replicate (bottom); columns are the true group labels, the k-means partition and the Ward partition. In the two partition columns, colours are matched across rows by Hungarian alignment, so the same colour denotes corresponding groups in the original and synthetic panels. The lower-left panel is drawn in a single colour: the group label in the synthetic data is itself synthesised, so it is a prediction from the synthetic features rather than a planted truth, and it has no bijective correspondence with the original groups. Both rows contain the same number of observations ($N = 1000$); the synthetic panels can look sparser only because overlapping points coincide more often. Scenario: $k = 2$, $p = 2$, $\rho = 0$, $\sigma = 2$.

instead with the number a clustering algorithm recovers from the *original* data isolates the effect of synthesis, since the same algorithm and the same estimation rule are applied on both sides. We report both throughout this section, and use the second elsewhere in the paper. Figure 3 shows each against separation, stratified by the true number of groups.

Separation is the decisive factor under both references. Against the planted truth, agreement rises from 0.272 (k-means) and 0.296 (Ward) at $\sigma = 0.1$, through 0.304 and 0.307 at $\sigma = 2$, to 0.810 and 0.789 at $\sigma = 6$ and 0.954 and 0.950 at $\sigma = 10$. The transition is sharp: below $\sigma \approx 2$ agreement is low and dominated by the overlap of the groups themselves, above $\sigma \approx 6$ it is near-perfect.

Against the partition recovered from the original data the picture is markedly more favourable, and the gap is largest exactly where the groups overlap: 0.751 (k-means) and 0.676 (Ward) at $\sigma = 0.1$, rising to 0.983 and 0.976 at $\sigma = 10$, for overall agreement of 0.864 and 0.823. The explanation is that at low separation the algorithms frequently fail to recover the planted count from the original data as well: at $\sigma = 0.1$, k-means finds the true k in only 0.289 of original datasets and Ward in 0.306. Much of what

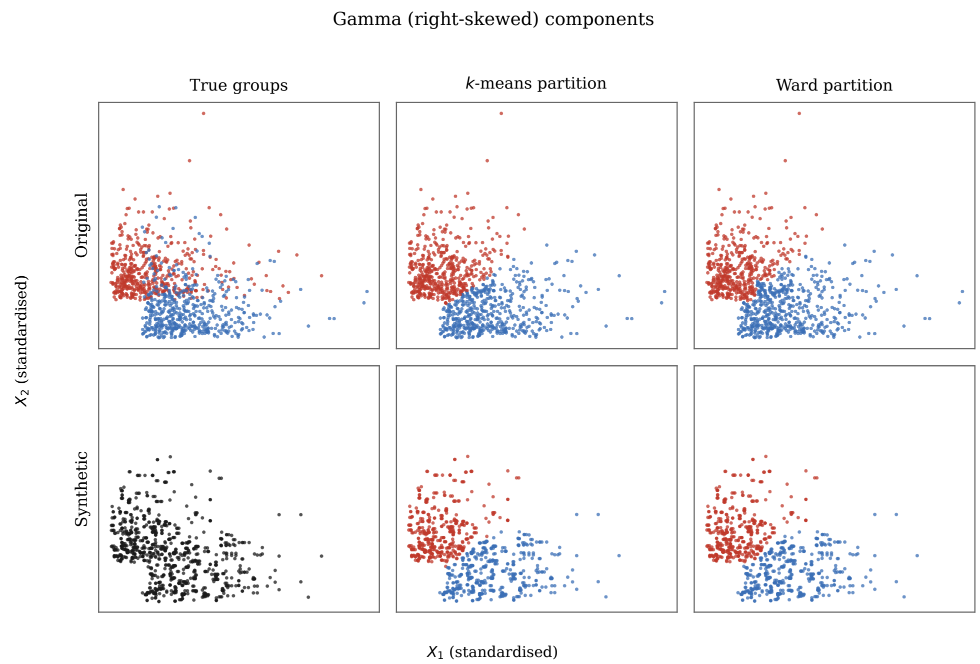

Fig. 2 Gamma (right-skewed) components: the same synthesis mechanism hardens smooth tails into axis-aligned blocks. Panel layout, colour matching, the single-colour lower-left panel and the scenario are as in Fig. 1, and both rows again contain $N = 1000$ observations. Comparing the lower-left panel with the upper-left one shows the artifact directly: the continuous density gradient of the original becomes a set of discrete rectangular patches.

the first reference records as a failure of synthesis is therefore a failure of clustering under overlap, which synthesis reproduces faithfully.

Recovery is also far easier for fewer groups. At $k = 2$ the mean recovery rate is 0.843 for k-means, against 0.473 at $k = 3$ and 0.439 at $k = 4$. The interaction is stark: in the overlapping regime the three- and four-group cases are almost never recovered, yet the same cases exceed 0.90 once $\sigma = 10$. Table 2 reports the marginal means for every design factor. Dimensionality and correlation are secondary, and the distribution family produces a consistent but modest gap that Section 3.4 examines directly.

3.3 Geometric fidelity

Figure 4 traces the two geometric-fidelity metrics against separation. Mean centroid distance falls steeply as groups separate, from 0.60 (k-means) and 0.86 (Ward) at $\sigma = 0.1$ to 0.09 and 0.10 at $\sigma = 10$, because well-separated groups pin their centroids robustly regardless of the local block artifact. Mean variance difference declines in parallel, from 0.35 and 0.61 to 0.11 and 0.13. At every separation level and on both metrics, k-means sits below Ward: its centroid averaging makes it the geometrically more conservative estimator under synthesis. The intervals overlap at $\sigma \geq 6$, where both methods are close to the floor.

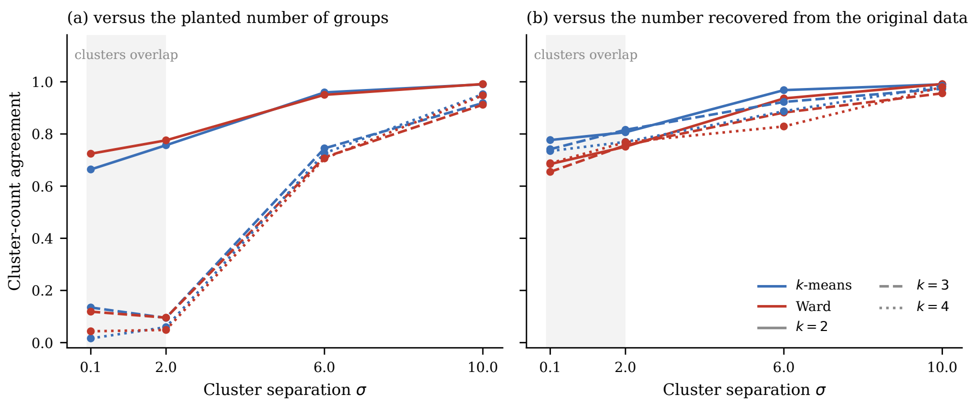

Fig. 3 Cluster-count agreement between synthetic and original data, against separation and stratified by the true group count, under two reference conventions. (a) Agreement with the number of groups planted by the simulation. (b) Agreement with the number a clustering algorithm recovers from the original data. Panel (a) is flat and low while the groups overlap (shaded), and collapses almost completely for $k \geq 3$; panel (b) stays high throughout and depends far less on k, because in the overlapping regime the algorithms often fail to recover the planted count from the original data either, so both sides of the comparison are wrong in the same way. The two algorithms track each other closely under (a) and separate slightly under (b). Both panels use all 720,000 original–synthetic pairs; the unit of replication is the pair.

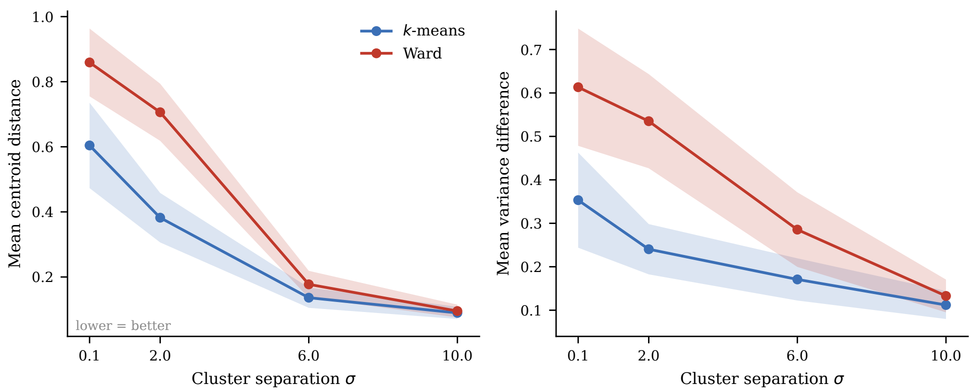

Fig. 4 Geometric fidelity against separation. Points are means over the 144 scenario-level means; bands are 95% intervals computed across scenarios. Left: mean centroid distance. Right: mean variance difference. Both decline as groups separate, and k-means remains below Ward throughout.

Dimensionality is the strongest amplifier of variance distortion. Figure 5(a) shows MVD growing steeply with the number of variables: for k-means from 0.06 at $p = 2$ to 0.39 at $p = 10$, and for Ward from 0.13 to 0.65. Each additional synthesised dimension compounds the axis-aligned leaf structure of Eq. (2). This is the mechanism behind the mild degradation of recovery at higher p in Table 2.

Figure 5(b) shows the group-balance error ΔG by distribution family. Both algorithms preserve balance better on Normal than on Gamma data (k-means 0.028 versus

Table 2 Mean recovery rate of the true number of groups by design factor, over the full 720,000-evaluation design.

Factor	Level	k-means	Ward
Separation σ	0.1	0.272	0.296
	2	0.304	0.307
	6	0.810	0.789
	10	0.954	0.950
Number of groups k	2	0.843	0.860
	3	0.473	0.459
	4	0.439	0.437
Dimensionality p	2	0.616	0.623
	5	0.561	0.569
	10	0.578	0.564
Correlation ρ	0	0.571	0.578
	0.4	0.599	0.592
Distribution	Normal	0.604	0.610
	Gamma	0.565	0.561
Overall		0.585	0.585

0.032; Ward 0.057 versus 0.065), consistent with the block artifact being more disruptive on skewed groups. More striking is that k-means yields systematically lower and tighter ΔG than Ward at both distributions. This reflects a structural property of the algorithm rather than better fidelity: the within-cluster sum-of-squares objective of Eq. (A1) favours partitions of comparable size, so k-means tends to equalise group sizes and thereby suppresses the natural imbalance that skewed data would otherwise produce. Ward imposes no such balancing pressure and follows the block-distorted density more literally, preserving imbalance at the cost of higher variance across replicates. This is the clearest algorithmic difference the study uncovers: not a difference in accuracy, but in the kind of geometric bias each method introduces.

3.4 Distribution family

The distribution family is the factor that exposes the CART artifact most directly. Figure 6 stratifies recovery by distribution and separation. In the overlapping regime the two families are indistinguishable, because recovery is limited by the overlap rather than by synthesis. In the informative regime the gap opens: at $\sigma = 6$, k-means recovers the Normal structure 86.2% of the time but the Gamma structure only 75.9% (Ward: 84.55% versus 73.3%). By $\sigma = 10$ the gap closes again, as separation overwhelms the artifact.

This addresses the first question. CART preserves the signal needed to recover the number of groups well on spherical Normal data, but its hardening of skewed densities measurably reduces recovery on Gamma data across precisely the range of separations where recovery is otherwise achievable.

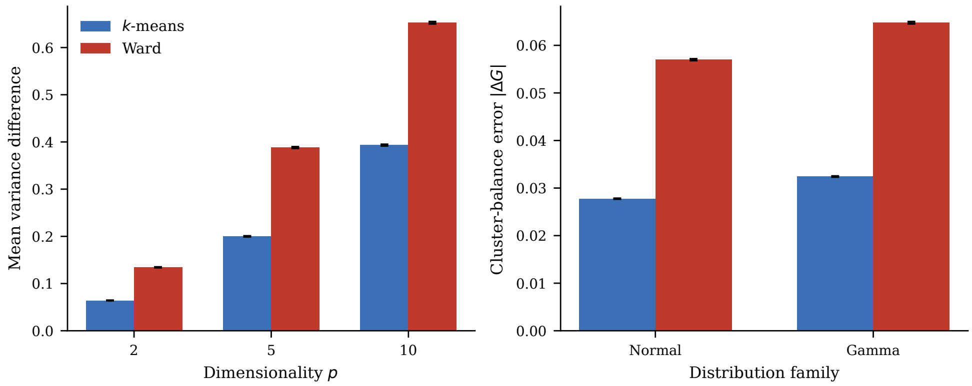

Fig. 5 (a) Mean variance difference by dimensionality: distortion grows steeply as dimensions are added, more so for Ward. (b) Group-balance error $|\Delta G|$ by distribution family: k-means produces lower and tighter balance error at both distributions, reflecting the balancing pressure of its objective function. Error bars are 95% intervals.

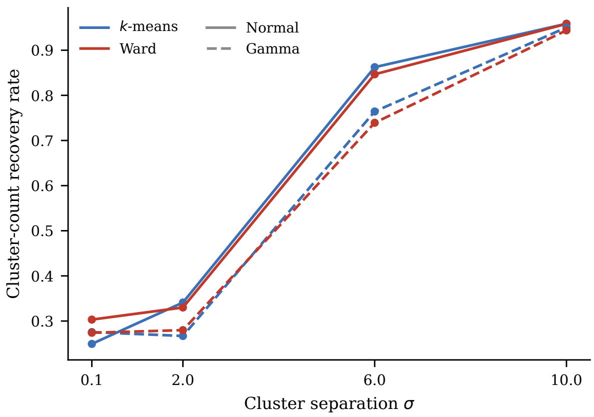

Fig. 6 Recovery of the number of groups by distribution family and separation. The families are indistinguishable while groups overlap and diverge in the informative regime, where Gamma data depresses recovery for both algorithms.

3.5 Agreement between the two algorithms

Whether the two paradigms differ in robustness to synthetic distortion was the third question. They do not, at least not on recovery. The overall recovery rates are 0.585 (k-means) and 0.585 (Ward); a paired Wilcoxon signed-rank test across the 144 scenario-level means finds no significant difference ($W = 3300.0$, $p = 0.39$), and the two algorithms agree on the outcome in 92.0% of individual evaluations. The hypothesis that connectivity-based clustering would be materially more robust (plausible given its ability to follow non-convex structure) is not supported.

As a complementary check on internal cohesion, the difference in mean silhouette width between synthetic and original partitions is small and centred near zero at every separation. The absolute levels are nearly identical (k-means 0.396 versus 0.397; Ward 0.376 versus 0.377), confirming that CART neither manufactures nor destroys apparent cluster quality. This is reassuring, but it also means that silhouette alone would fail to flag the geometric distortions the other metrics detect, an instructive negative result for anyone tempted to validate a synthetic release with an internal index.

4 Discussion

This study evaluated whether CART-generated synthetic data preserves the structural properties that cluster analysis relies on, comparing k-means and Ward's hierarchical clustering across a 144-scenario factorial design and 720,000 synthetic-versus-original evaluations. Four findings emerge.

First, separation dominates recovery of the number of groups. The recovery rate rises from roughly 0.27 in the overlapping regime to above 0.95 at $\sigma = 10$, with the number of groups the second factor: easy at $k = 2$ (0.84) and hard at $k \geq 3$ (below 0.48). Dimensionality, correlation and distribution family act as secondary modulators. A practical implication is that a synthetic release is most trustworthy for clustering exactly where clustering is easiest, and least trustworthy where the analyst most needs help.

Second, the two algorithms are statistically indistinguishable on recovery (0.585 versus 0.585; $p = 0.39$; 92% agreement). Where they differ is in the geometric bias they introduce. k-means is the more conservative estimator (lower centroid and variance distortion at every separation), and its objective imposes a balancing bias that suppresses group-size imbalance. Ward preserves imbalance more faithfully but with higher variance. For an applied analyst this is the more actionable result than the null on accuracy: if group sizes are themselves a finding, a k-means analysis of synthetic data will understate imbalance in a way that is invisible from the partition alone.

Third, CART is not utility-neutral, and its failure mode is distributional. On spherical Normal data it is close to neutral; on skewed Gamma data it hardens smooth density gradients into axis-aligned blocks, visibly distorting geometry and measurably depressing recovery in the informative regime. Degradation grows with dimensionality, as each synthesised dimension compounds the block structure.

Fourth, and methodologically, an internal validity index is not a sufficient check. Silhouette geometry is preserved almost exactly even where centroid and dispersion fidelity degrade substantially. A practitioner who validates a synthetic release by confirming that its silhouette profile matches the original will conclude that structure is intact in precisely the settings where this analysis shows it is not.

An interesting corollary of the design is that clustering can be read in the other direction, as an instrument rather than an object of study. Because a partition responds to local density, separability and dispersion simultaneously, the agreement between partitions recovered from original and synthetic data functions as a sensitive summary of geometric resemblance, one that reacts to distortions which marginal and bivariate checks do not see. We did not pursue this here, but the metrics of Section 2.4 could be repurposed as general-utility diagnostics rather than task-specific ones.

4.1 Relation to previous work

The study extends Chen (2025) in three ways: a structurally distinct second algorithm, a skewed distribution family that exposes tree-synthesis artifacts, and ground-truth-referenced structural metrics that do not depend on the synthetic data alone. The conclusion that CART is a reasonable default for well-behaved data but degrades on skewed distributions is consistent with the broader `synthpop` literature (Nowok et al, 2016; Snoke et al, 2018), and the qualitative block artifact matches the known axis-aligned bias of sequential tree synthesis. The finding that utility rankings depend on the downstream task echoes benchmarking work in the health-records setting (Yan et al, 2022; El Emam et al, 2022), where the same generator changes position depending on which analysis is used to score it. A parallel question arises when values are missing rather than synthesised: Aschenbruck et al (2023) evaluate how imputation strategies affect the clusters subsequently recovered from mixed-type data, and reach a structurally similar conclusion, that the imputation model and the clustering method interact and cannot be assessed in isolation. Since CART underlies both sequential synthesis and widely used non-parametric imputation, the two literatures are closer than their vocabularies suggest.

4.2 Limitations

The evaluation uses controlled simulations with known ground truth rather than real administrative data. This isolates mechanisms cleanly but limits direct generalisation to messy data with missingness, mixed types and unknown data-generating processes. The generator is limited to CART and the algorithms to k-means and Ward; density-based and model-based clustering, and parametric or deep generative synthesisers, may behave differently. Sample size was fixed at $N = 1000$, so small-sample behaviour is unexamined, and the geometric-fidelity metrics were computed on a balanced subset of the synthetic draws rather than all of them.

Two conventions in the definition of MCD and MVD deserve emphasis, because both affect the numbers themselves and not only their interpretation. The first is what the metrics compare. Here they compare the partition recovered from the original data against the partition recovered from the synthetic data, which is the quantity a claim about synthesis requires; the superficially similar practice of comparing a recovered partition against the planted labels of the same dataset measures clustering accuracy instead, and carries no information about the generator. The second is scale. Both metrics are computed after standardising with a scaler fit on the original data and applied unchanged to the synthetic data, so both are expressed in units of the original feature scale. Standardising each dataset separately rescales away part of the distortion the metrics are meant to capture, and omitting standardisation altogether makes MVD track absolute cluster spread (which grows with σ by construction), so that the metric rises with separation when what it is measuring is the design parameter. Absolute values of MCD and MVD are therefore not comparable across studies that differ in either convention, and any replication should state both explicitly.

4.3 Future work

The central lesson generalises beyond the specific methods studied: synthetic data utility is task-relative, and a generator that preserves marginal resemblance need not preserve the higher-order geometry a clustering workflow depends on. Practically, the results argue for validating structural fidelity (distances, dispersion, separability, balance) against the intended analysis, and for treating a clustering comparison as the more demanding resemblance check that marginal summaries cannot provide. Natural extensions include other synthesisers (parametric mixtures, copulas, deep generative models), non-convex and density-based clustering, varying N to probe small-sample behaviour, and real datasets with established latent structure. A promising direction is to model the discrepancy Δ of Eq. (1) directly as a learned function of the design factors, turning the utility surface itself into a predictive tool.

Code availability

The full pipeline, from data generation through to the figures and tables reported here, is available at https://github.com/Sterrysx/synthetic-clustering. Every stage is driven from a single configuration file and can be re-executed end to end.

Synthetic data generated with CART can reproduce the appearance of clustering structure while subtly reshaping its geometry. Both algorithms recover well-separated structure reliably and fail together when groups overlap; where they diverge is in the bias each imprints on the recovered partition. The utility of synthetic data for clustering is therefore best understood not as a property of the data but as a property of the data–generator–algorithm triple, to be verified rather than assumed.

Appendix A Clustering objectives

k-means partitions N observations $\{x_i\}_{i=1}^N$, $x_i \in \mathbb{R}^p$, into k non-overlapping groups by minimising the within-cluster sum of squares (MacQueen, 1967). Writing $\mathcal{C} = \{C_1, \ldots, C_k\}$ for the partition and μ_j for the centroid of C_j,

$$\arg\min_{\mathcal{C}} \sum_{j=1}^{k} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2, \qquad \mu_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i. \tag{A1}$$

Lloyd's iteration alternates assignment and update steps until the partition stabilises, converging to a local optimum.

Hierarchical clustering builds a dendrogram by successive merging (Ward, 1963). Each observation begins as a singleton; at every step the two clusters whose merger least increases the total within-cluster error sum of squares are combined. With

$$\mathrm{ESS} = \sum_{j} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2, \qquad \Delta_{\mathrm{Ward}}(C_a, C_b) = \frac{|C_a|\,|C_b|}{|C_a| + |C_b|} \|\mu_a - \mu_b\|^2, \tag{A2}$$

a flat partition into k groups is obtained by cutting the dendrogram at the corresponding height.

Appendix B Computational complexity and cost

This appendix states what the study costs to run and why. Throughout, N is the sample size, p the dimensionality, k the number of groups, I the number of Lloyd iterations to convergence, $K = 7$ the number of candidate values of $\hat{k}$, and $A = 2$ the number of algorithms.

B.1 Cost of the components

Table B1 collects the asymptotic cost of each component together with the number of times the executed design invokes it. The invocation counts follow from the design: 144 scenarios $\times$ $n = 5$ replicates $= 720$ units, each with $m = 1000$ synthetic draws, giving 720,000 original–synthetic pairs.

Table B1 Cost per invocation and invocation count. Exact minimisation of the k-means objective is NP-hard in general and remains so for $k = 2$ in arbitrary dimension (Inaba et al, 1994), so Lloyd's local-search heuristic (Lloyd, 1982) is used, restarted ten times.

Component	Cost per invocation	Invocations
CART synthesis (Breiman et al, 1984)	$O(p^2 N \log N)$	1.44×10^5
Lloyd iteration (MacQueen, 1967; Lloyd, 1982)	$O(IkNp)$, $\times 10$ restarts	5.77×10^6
Ward linkage (Ward, 1963; Murtagh and Contreras, 2017)	$O(N^2 p)$ time, $O(N^2)$ space	5.77×10^6
Silhouette (Rousseeuw, 1987)	$O(N^2 p)$	1.15×10^7
Hungarian alignment (Kuhn, 1955; Jonker and Volgenant, 1987)	$O(k^3)$	1.44×10^6

Two structural facts follow. First, the *binding* term is $O(N^2 p)$, contributed by Ward's linkage and by the silhouette, and it is a property of the sample size rather than of the design: doubling N quadruples the work, whereas doubling m or the number of scenarios only doubles it. The $O(N^2)$ working set of the linkage is also what caps N in practice, since each worker holds its own distance structure; at $N = 1000$ this is roughly 8 MB per worker and irrelevant, at $N = 10^4$ it is 800 MB and binding. Second, model selection dominates the fit count: estimating $\hat{k}$ costs K fits and K silhouette evaluations per algorithm per dataset, so the fidelity stage performs $720 \times A(K+1) + 720{,}000 \times A(K+1) = 1.15 \times 10^7$ clustering fits rather than the 1.44×10^6 that fitting at a single known k would require. Where a cheaper formulation exists we did not pursue it: Ward admits an $O(N^2)$ nearest-neighbour-chain implementation (Murtagh and Contreras, 2017; Müllner, 2011) and k-means can be seeded to bound its expected error (Arthur and Vassilvitskii, 2007), so the figures below are conservative relative to a tuned implementation.

One algorithmic choice does change the constant materially. The partition of each original dataset is computed once per replicate and reused across that replicate's 1000 draws. Computing it inside the draw loop would multiply the original-side work by 10^3 and produce identical output, since the original data do not vary within a replicate.

B.2 Parallel execution

The design is a set of independent units, so the computation is embarrassingly parallel in the strict sense: no communication, no shared mutable state, and a serial fraction confined to reading configuration and writing the merged result. Amdahl's law is therefore not the operative constraint at this scale; hardware contention is, and the implementation targets three specific sources of it.

Thread oversubscription. The clustering routines call into BLAS, which by default spawns one thread per core *inside each worker*. With P worker processes this creates $P \times C$ runnable threads on C cores, and the resulting context-switching and cache thrashing dominate the useful work. All three stages therefore pin `OMP_NUM_THREADS`, `OPENBLAS_NUM_THREADS` and `MKL_NUM_THREADS` to 1 before importing any numerical library, leaving parallelism entirely to the process pool. The effect is not marginal: on the reference machine, 18 concurrent workers each running an identical model-selection scan complete in 1.9 s with threads pinned and 84.5 s without, a factor of 44.

Load imbalance. Unit cost varies with p and k, so a static partition of the task list leaves workers idle at the tail. The synthesis stage instead uses a work-stealing queue: tasks are serialised into a spool directory and each worker claims one by `rename(2)` into a *doing* directory. Rename within a filesystem is atomic, so exactly one worker can win a given task and no lock is required. Workers pull candidates in batches of 200 rather than listing the spool per task, which at 1.44×10^5 tasks reduces directory-scan overhead from hours to seconds. The Python stages achieve the same effect through a dynamically scheduled pool.

Restart cost. Each unit checkpoints its own result on completion, so an interrupted run resumes at the last completed unit rather than from the start. At a stage cost of several hours this converts a failure from a total loss into the loss of the units in flight.

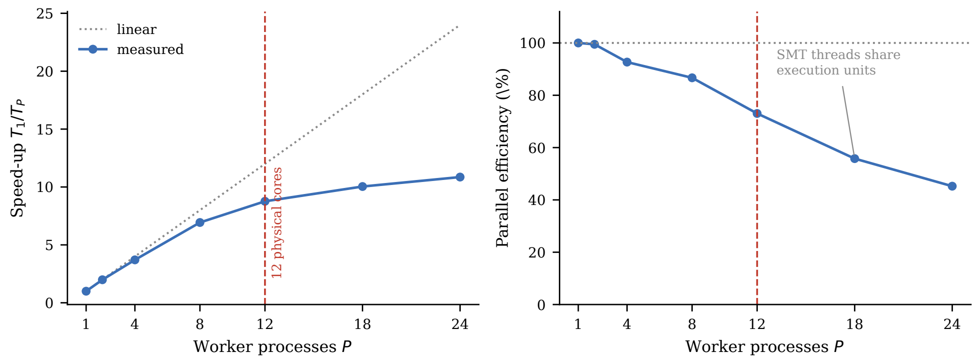

Fig. B1 Strong scaling of the model-selection workload on the reference machine, 72 independent units, BLAS pinned to one thread per worker. Scaling is near-linear to 8 workers (6.9×, 87% efficiency) and the knee falls at 12, the physical core count. Beyond it the additional workers occupy SMT siblings that share execution units with an already-saturated core, so throughput still rises but efficiency falls to 56% at 18 workers and 45% at 24. The production runs use 18: past the efficiency knee, but chosen for wall time on an otherwise idle machine. The efficiency figure is not a curiosity: at 56%, the clustering stage costs 3.6 h rather than the 2.0 h that dividing its serial cost by 18 would suggest.

Figure B1 shows the measured strong-scaling curve. The shape is the expected one for a compute-bound, memory-light workload on a simultaneous multithreading processor, and it makes the operating point explicit: 18 workers buy a 10.0× speed-up over serial execution, while 12 would give 8.8× at much better efficiency. Neither choice is a contribution; the point of measuring is that the default of one worker per hardware thread is not obviously right, and here it is not.

B.3 Measured cost

Table B2 Wall-clock cost at $m = 1000$ on an AMD Ryzen 9 9900X (12 cores, 24 threads, 46 GiB) with 18 workers. The evidence differs by row and is stated rather than blurred: the fidelity stage is instrumented; synthesis is the interval over which its 1.44×10^5 outputs were written, so it includes any idle time inside that interval; the clustering stage was not instrumented, so its entry is a measurement on a uniform random sample of 3000 of its 1.44×10^5 inputs, run at the same worker count, scaled to the full set; this is consistent with the loose upper bound of 5.6 h implied by its output timestamps.

Stage	Evidence	Wall time	Unit cost
Synthesis (1.44×10^5 datasets)	write interval	1.24 h	31 ms/dataset
Clustering (7.2×10^5 pairs)	sampled	$\approx$ 3.6 h	$\approx$ 18 ms/pair
Fidelity (1.15×10^7 fits)	instrumented	4.44 h	1.4 ms/fit
Total		$\approx$ 9.3 h	

The synthetic datasets occupy 21 GB as uncompressed Parquet. Cost is linear in m: the same design at $m = 100$ completes in about a tenth of the time and reproduces every estimate reported here to within 0.025, so the larger run buys precision rather than a change in any conclusion.

References

Ahmed M, Seraj R, Islam SMS (2020) The k-means algorithm: A comprehensive survey and performance evaluation. Electronics 9(8):1295. https://doi.org/10.3390/electronics9081295

Arthur D, Vassilvitskii S (2007) *k*-means++: The advantages of careful seeding. In: Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms, pp 1027–1035

Aschenbruck R, Szepannek G, Wilhelm AFX (2023) Imputation strategies for clustering mixed-type data with missing values. Journal of Classification 40(1):2–24. https://doi.org/10.1007/s00357-022-09422-y

Bhanot K, Qi M, Erickson JS, et al (2021) The problem of fairness in synthetic healthcare data. Entropy 23(9):1165. https://doi.org/10.3390/e23091165

Breiman L, Friedman JH, Olshen RA, et al (1984) Classification and Regression Trees. Wadsworth, Belmont, CA

Chen X (2025) Evaluating the utility of synthetic data for clustering. Bachelor's thesis, Universitat Politècnica de Catalunya, Barcelona

Drechsler J (2011) Synthetic Datasets for Statistical Disclosure Control: Theory and Implementation. Springer, New York

El Emam K, Mosquera L, Bass J (2020) Evaluating identity disclosure risk in fully synthetic health data: Model development and validation. Journal of Medical Internet Research 22(11):e23139. https://doi.org/10.2196/23139

El Emam K, Mosquera L, Fang X, et al (2022) Utility metrics for evaluating synthetic health data generation methods: Validation study. JMIR Medical Informatics 10(4):e35734. https://doi.org/10.2196/35734

Giuffrè M, Shung DL (2023) Harnessing the power of synthetic data in healthcare: Innovation, application, and privacy. npj Digital Medicine 6:186. https://doi.org/10.1038/s41746-023-00927-3

Gonçalves A, Ray P, Soper B, et al (2020) Generation and evaluation of synthetic patient data. BMC Medical Research Methodology 20:108. https://doi.org/10.1186/s12874-020-00977-1

Gonzales A, Guruswamy G, Smith SR (2023) Synthetic data in health care: A narrative review. PLOS Digital Health 2(1):e0000082. https://doi.org/10.1371/journal.pdig.0000082

Goyal M, Mahmoud QH (2024) A systematic review of synthetic data generation techniques using generative AI. Electronics 13(17):3509. https://doi.org/10.3390/electronics13173509

Inaba M, Katoh N, Imai H (1994) Applications of weighted Voronoi diagrams and randomization to variance-based k-clustering. In: Proceedings of the Tenth Annual Symposium on Computational Geometry, pp 332–339, https://doi.org/10.1145/177424.178042

Jonker R, Volgenant A (1987) A shortest augmenting path algorithm for dense and sparse linear assignment problems. Computing 38(4):325–340. https://doi.org/10.1007/BF02278710

Kokosi T, De Stavola B, Mitra R, et al (2022) An overview on synthetic administrative data for research. International Journal for Population Data Science 7(1):1727. https://doi.org/10.23889/ijpds.v7i1.1727

Kuhn HW (1955) The Hungarian method for the assignment problem. Naval Research Logistics Quarterly 2(1–2):83–97

Lautrup AD, Hyrup T, Zimek A, et al (2024) Systematic review of generative modelling tools and utility metrics for fully synthetic tabular data. ACM Computing Surveys 57(4):1–38. https://doi.org/10.1145/3704437

Little RJA (1993) Statistical analysis of masked data. Journal of Official Statistics 9(2):407–426

Liu X, Song W, Wong BY, et al (2019) A comparison framework and guideline of clustering methods for mass cytometry data. Genome Biology 20:297. https://doi.org/10.1186/s13059-019-1917-7

Lloyd SP (1982) Least squares quantization in PCM. IEEE Transactions on Information Theory 28(2):129–137. https://doi.org/10.1109/TIT.1982.1056489

MacQueen J (1967) Some methods for classification and analysis of multivariate observations. Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability 1:281–297

Matthews GJ, Harel O (2011) Data confidentiality: A review of methods for statistical disclosure limitation and methods for assessing privacy. Statistics Surveys 5. https://doi.org/10.1214/11-SS074

Müllner D (2011) Modern hierarchical, agglomerative clustering algorithms. arXiv preprint arXiv:11092378 https://doi.org/10.48550/arXiv.1109.2378

Murtagh F, Contreras P (2017) Algorithms for hierarchical clustering: An overview, II. WIREs Data Mining and Knowledge Discovery 7(6). https://doi.org/10.1002/widm.1219

Nowok B, Raab GM, Dibben C (2016) synthpop: Bespoke creation of synthetic data in R. Journal of Statistical Software 74(11). https://doi.org/10.18637/jss.v074.i11

Pedregosa F, et al (2011) Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12:2825–2830

Quintana DS (2020) A synthetic dataset primer for the biobehavioural sciences to promote reproducibility and hypothesis generation. eLife 9:e53275. https://doi.org/10.7554/eLife.53275

R Core Team (2024) R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria, URL https://www.R-project.org/, version 4.3

Rankin D, Black M, Bond R, et al (2020) Reliability of supervised machine learning using synthetic data in health care: Model to preserve privacy for data sharing.

JMIR Medical Informatics 8(7):e18910. https://doi.org/10.2196/18910

Reiner Benaim A, Almog R, Gorelik Y, et al (2020) Analyzing medical research results based on synthetic data and their relation to real data results: Systematic comparison from five observational studies. JMIR Medical Informatics 8(2):e16492. https://doi.org/10.2196/16492

Rousseeuw PJ (1987) Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics 20:53–65. https://doi.org/10.1016/0377-0427(87)90125-7

Rubin DB (1993) Statistical disclosure limitation. Journal of Official Statistics 9(2):461–468

Snoke J, Raab GM, Nowok B, et al (2018) General and specific utility measures for synthetic data. Journal of the Royal Statistical Society: Series A (Statistics in Society) 181(3):663–688. https://doi.org/10.1111/rssa.12358

Templ M, Meindl B, Kowarik A, et al (2017) Simulation of synthetic complex data: The R package simPop. Journal of Statistical Software 79(10). https://doi.org/10.18637/jss.v079.i10

Ward JH (1963) Hierarchical grouping to optimize an objective function. Journal of the American Statistical Association 58(301):236–244. https://doi.org/10.2307/2282967

Yan C, Yan Y, Wan Z, et al (2022) A multifaceted benchmarking of synthetic electronic health record generation models. Nature Communications 13:7609. https://doi.org/10.1038/s41467-022-35295-1

Zappia L, Oshlack A (2018) Clustering trees: A visualization for evaluating clusterings at multiple resolutions. GigaScience 7(7):giy083. https://doi.org/10.1093/gigascience/giy083

Zellinger MJ, Bühlmann P (2025) Natural language-based synthetic data generation for cluster analysis. Journal of Classification 42(3):517–543. https://doi.org/10.1007/s00357-025-09501-w

Zhao Z, Kunar A, Birke R, et al (2024) CTAB-GAN+: Enhancing tabular data synthesis. Frontiers in Big Data 6:1296508. https://doi.org/10.3389/fdata.2023.1296508